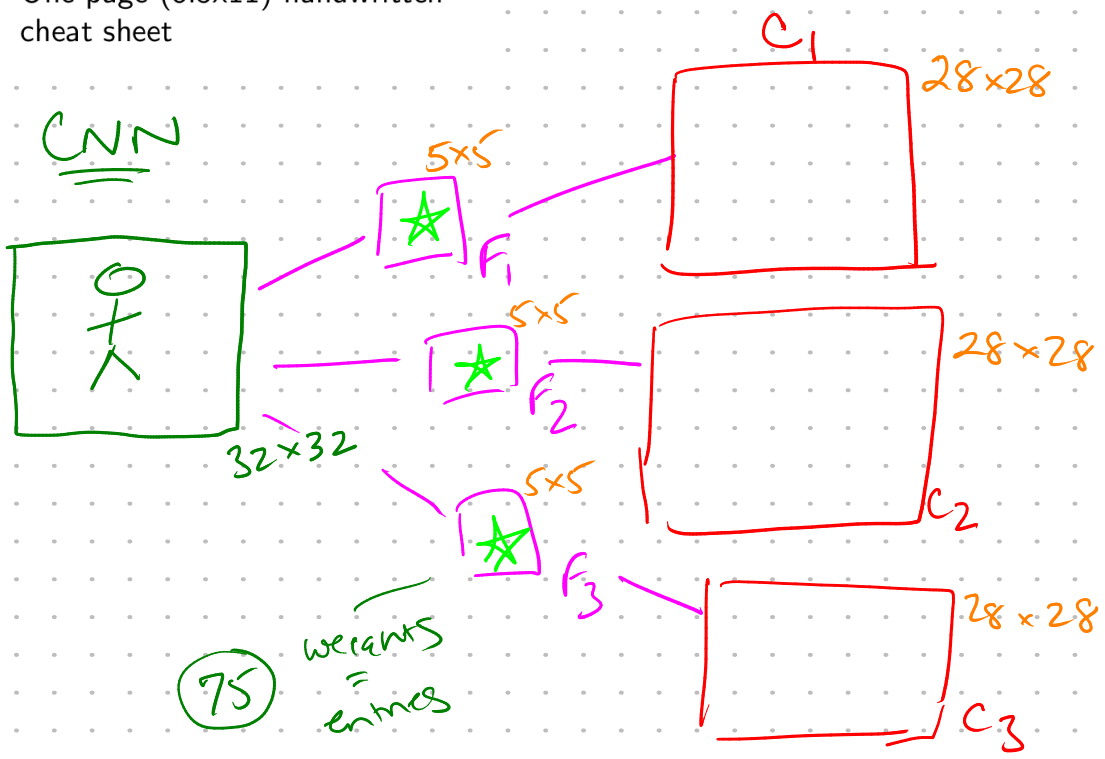
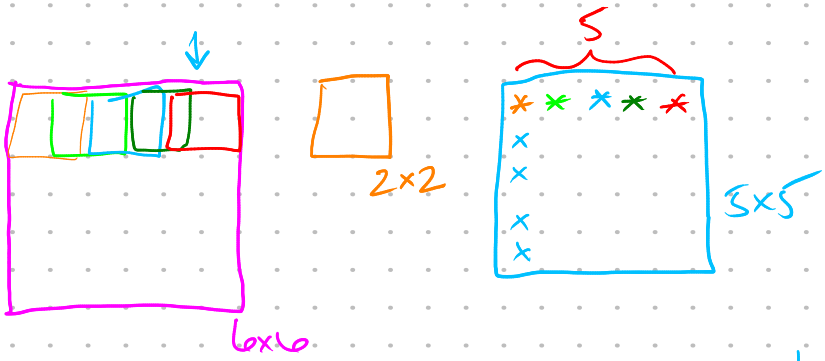
This week:

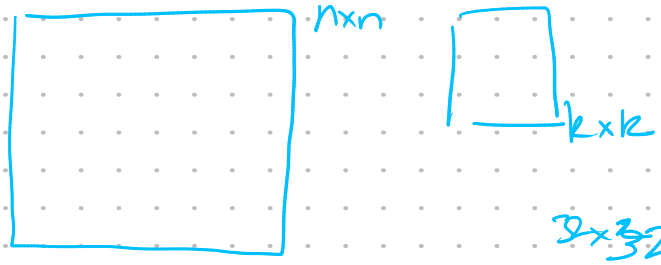
- Mon 4/25: 12.1, 12.4: Clustering
- Weds 4/27: Finish 12.4 & Mini lab
  - HW #10 Due Weds
- Fri 4/29: Review - Bring questions!
- Final: May 4, 5:45 - 7:45 PM, Erickson 226
  - Content since midterm (Ch 6 and on)
  - One page (8.5x11) handwritten cheat sheet





$6-2+1=5$

$n-k+1$



32×32

$32-5+1=28$

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

$X = (x_1, x_2, \ldots, x_p)$

Lin $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$

not the same

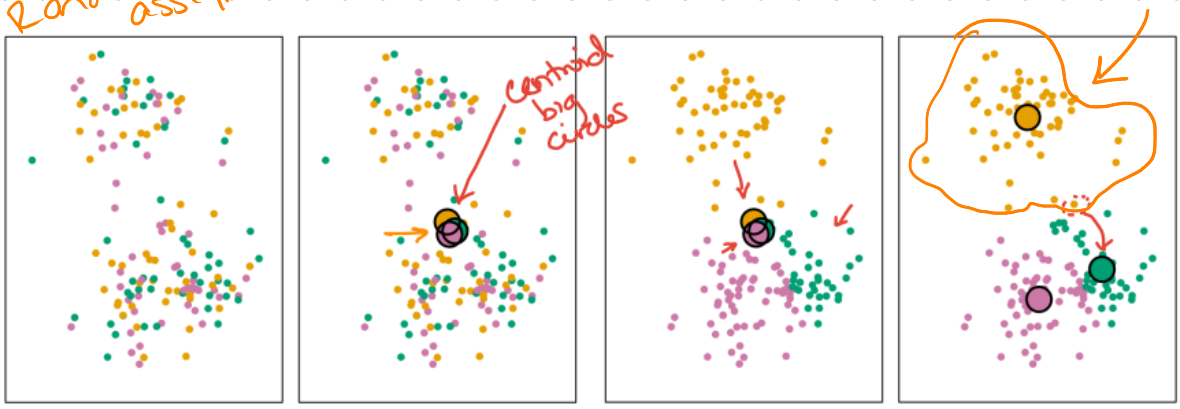
Input = single number x_i ← i^{th} data point

$\beta_0 + \beta_1 mpg + \beta_2 mpg^2 + \beta_3 mpg^3$

$$X = \begin{array}{c} \\ \text{cars} \end{array} \begin{array}{c} mpg \\ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \end{array} \qquad \beta_0 + \beta_1 (mpg)$$

$$\begin{array}{ccc} mpg & mpg^2 & mpg^3 \\ \end{array}$$
$$\begin{array}{c} \\ \text{orig} \end{array} \begin{pmatrix} | & | & | \\ | & | & | \end{pmatrix}$$

Random assignment



orange

just orange

$$\left(\frac{1}{|C_{or}|}\left[\sum_{i \in C_{orange}} (x_{i,1})\right], \frac{1}{|C_{or}|}\left[\sum_{i \in C_{or}} (x_{i,2})\right] \right)$$